07 The Predator Room

Predator–prey cycles

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 36 min	**+ 28 min**	The Crowding (02) and Resource (03) rooms. The Overshoot Room (04) gives a second kind of cycle to compare.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Predator tab

Two species, one of which eats the other. Three versions of the same pair of equations each repair an assumption of the one before: classic Lotka–Volterra, prey with a carrying capacity, and a predator that takes time to handle each prey. This guide asks why the pair cycles, and what makes the cycles grow, shrink or settle.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain why predator peaks lag behind prey peaks.
- Distinguish neutral cycles (size set by the starting point) from a stable equilibrium and from a limit cycle.
- Show how giving the prey a carrying capacity damps the cycles.
- Describe the paradox of enrichment.

The extension adds

- Use $N^* = m/(a(e - mh))$ to show that the prey equilibrium contains no prey parameters.
- Predict the critical K above which a saturating predator destabilises the system.
- Explain why large classic cycles can drive a population to extinction.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Two-room comparison:** this room with the Overshoot Room, through question 3.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Neutral cycles

10 min

SET UP

Preset **Neutral cycles** (classic Lotka–Volterra).

PREDICT FIRST

Will the cycles grow, shrink, or stay the same size? If you start from a different point, will you get the same cycle?

RUN

Run. Then click several different points on the phase plane.

LOOK AT

the orbits on *Phase plane & nullclines*.

EXPLAIN

Why is the cycle you get the cycle you started with? Why is that biologically implausible?

2 The lag

8 min

SET UP

The same preset.

PREDICT FIRST

Which peaks first, prey or predator? By how much of a cycle?

RUN

Run and look at the small dots that mark the peaks.

EXPLAIN

Explain the lag step by step: what happens to each species during one turn of the cycle?

3 Prey with a ceiling

10 min

SET UP

Preset **Damped to equilibrium** (prey with a ceiling, $K = 300$).

PREDICT FIRST

Click the same starting points as in step 1. Will the runs still keep their own cycles?

RUN

Run, then click several points on the phase plane.

EXPLAIN

Why does prey self-limitation pull every run to the same equilibrium?

4 The paradox of enrichment

8 min

SET UP

Preset **Paradox of enrichment** (predator that gets full, $K = 400$).

PREDICT FIRST

Will enriching the prey's environment (raising K) make the system more stable or less? Drag K down slowly: when will the cycling stop?

RUN

Run, then lower K step by step, running each time.

EXPLAIN

Why can enrichment destabilise a predator–prey pair?

EXTENSION · MSC

5 Who controls whom

10 min

SET UP

Prey with a ceiling. Change the prey's r and K; then change the predator's m.

PREDICT FIRST

Which of these changes will move the prey equilibrium N? *Which will move the predator equilibrium* P? Calculate N^* from $N^* = m/(a(e - mh))$ for the predator that gets full, with the default parameters.

RUN

Run after each change and read *Predicted vs. measured*.

EXPLAIN

Why is the prey's abundance set by the predator's biology? What does that imply for a trophic cascade?

6 The critical K

10 min

SET UP

Predator that gets full, defaults ($a = 0.02$, $h = 0.20$, $e = 0.40$, $m = 0.30$).

PREDICT FIRST

The equilibrium is stable when it lies to the right of the hump of the prey nullcline, which peaks at $(K - 1/(ah))/2$. Calculate the K at which stability is lost.

RUN

Run at K just below and just above your value.

EXPLAIN

Explain the result graphically, using where the predator nullcline crosses the hump.

7 Cycles that crash

8 min

SET UP

Preset **Cycles that crash** (classic, prey start at 250, predators at 8).

PREDICT FIRST

Which species will fall below half an individual and be lost, and why?

RUN

Run, using a logarithmic axis.

EXPLAIN

Why are large neutral cycles dangerous for a real population? What does the room's half-individual floor stand for?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc The Canadian lynx and snowshoe hare records show regular cycles. List two assumptions of this room that the real system breaks.
2. BSc Why would a predator with alternative prey be less likely to drive its prey to extinction?
3. BSc Compare the cycles here with those of the Overshoot Room. What causes each?
4. MSc Rosenzweig (1971) named the paradox of enrichment. Why has it been hard to demonstrate in the field?
5. MSc Explain the difference between a neutral cycle and a limit cycle in terms of what happens after a small perturbation.
6. MSc How could prey refuges stabilise a predator–prey system?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The cycles stay the same size; each starting point has its own orbit. Real populations are constantly perturbed, so this would make their cycles wander without limit.
- **Step 2.** Prey peak first; predators peak about a quarter of a cycle later. Prey grow while predators are few; predators then build up, eat the prey down, and starve.
- **Step 3.** Every start spirals into the same equilibrium: prey self-limitation damps the oscillation.
- **Step 4.** At $K = 400$ the system settles into a limit cycle; lowering K below about 340 brings it back to a stable equilibrium. Enrichment destabilises.
- **Step 5.** Prey r and K move only P*; predator parameters move* N. With handling, $N^* = 0.3/(0.02 \times (0.4 - 0.06)) \approx 44$.
- **Step 6.** The hump peaks at $(K - 250)/2$; stability is lost when that exceeds 44, so at $K \approx 338$.
- **Step 7.** The orbit is so large that the prey are eaten down below half an individual within the first few time units and are lost; with nothing to eat, the predators then dwindle to extinction too. The floor stands for the fact that real populations come in whole individuals.